

SATVIK TAJANE

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EDUCATION

Northeastern University, Boston, MA

December 2027

Candidate for Master of Science in Electrical and Computer Engineering

Relevant Courses: Mobile Robotics, Autonomous Field Robotics, Computer Vision, Machine Learning

MIT World Peace University, Pune, India

August 2025

Bachelor of Technology in Electrical and Computer Engineering

Relevant Courses: Artificial Intelligence, Data Structures and Algorithms, Robotics, Data Science for Engineers

SKILLS

- **Programming:** Python, C++
- **Robotics Frameworks:** ROS2, Gazebo, Moveit, Nav2, Rviz
- **Libraries:** OpenCV, Open3D, PyTorch, GTSAM
- **Computer Vision:** Visual Odometry, ArUco, Point Clouds
- **Development tools:** Gazebo, Issac Sim, Linux, Git, GitLab, Arduino, Raspberry pi

ACADEMIC PROJECTS

Robrewto | Vision Guided Manipulator

April 2026 - July 2026

- Architected a manipulation pipeline in ROS2 and Moveit on a Universal Robotics UR3e arm for vision guided manipulation
- Implemented a camera to robot perception pipeline in Gazebo to extract dynamic 6-DOF pose using ArUco markers and solvePNP with **low reprojection error of 0.16px**
- Integrated a hybrid motion planning framework combining OMPL RRTConnect for global collision avoidance with deterministic Cartesian path interpolation for near field pick execution

Stereo Visual SLAM with Pose Graph Optimization

April 2026 - May 2026

- Engineered a stereo visual odometry pipeline on the KITTI dataset using OpenCV, deriving metric depth from disparity to eliminate scale ambiguity
- Instantiated a GTSAM pose graph slam backend with Levenberg-Marquardt optimization, constructing relative 3D pose factors and an anchored prior to bound unbounded odometric drift over a 3.7km trajectory.
- Configured a descriptor based loop closure model with geometric verification, achieving a **92% drift reduction** and a 3.3 m RMSE ATE

Fault Tolerant Multimodal Localization System

January 2026 - April 2026

- Developed a dynamic state estimation framework integrating sensor fusion, including IMU, monocular camera, and ultrasonic sensor deployed on a custom Gazebo rover
- Formulated a real time sensor sensor reweighting via likelihood tempering for sensor influence based on a live confidence monitor to achieve graceful degradation under sensor dropouts
- Constructed a ROS2 fault injection pipeline for sensor simulation to evaluate performance and outperformed the Nav2 AMCL baseline by 1m ATE under severe weather

Real Time Physics Informed Motion Predictor for Autonomous Systems

September 2026 - December 2026

- Designed a physics informed linear transformer with $O(n)$ attention complexity for short horizon robot trajectory prediction
- Implemented a constrained output layer predicting control inputs instead of directly regressing Pose, integrated through a unicycle kinematic model to guarantee physically consistent trajectories
- Achieved ~2.36ms inference latency, 40× faster than a 10Hz budget on the NCLT dataset, demonstrating feasibility for real time onboard deployment

ACADEMIC PROJECTS

Alpha-ICT, Pune, India

January 2025 - June 2025

- Developed and deployed a Hardware-in-the-Loop (HIL) fault injection rig to simulate dynamic HVAC sensor dropouts
- Developed automated Python and Robot Framework test suites to execute multi parameter degradation loops on live sensor streams, accelerating testing by 45%